

DIRGHA JIVANI

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Education

Boston University

MS in Computer Science

August 2024 - Expected Jan 2026

Boston, Ma

Nirma University

B.Tech in Computer Science

Oct 2020 - June 2024

Ahmedabad, Gujarat

Industrial Experience

Software Engineering Intern

Cimcon Digital

Oct 2023 – May 2024

Ahmedabad

- Designed, trained, and deployed a real-time defect detection system using YOLO, Python, and OpenCV for Colgate's manufacturing line, achieving 92% detection accuracy and reducing manual QA workload by 40% through automated defect flagging, anomaly logging, and reporting workflows.
- Integrated a YOLO-based anomaly classification feature into the chatbot platform, enabling automatic responses to sensor video uploads and improving user experience.
- Automated weekly performance reports using Python and SQL, surfacing actionable insights on edge cases and model accuracy to guide product improvements.
- Built a RAG-based chatbot (LLM + SQL) into the client dashboard, reducing support response time by 30% and deflecting 20% of routine queries.

Data Science Intern

Tusker AI

June 2023 – July 2023

Ahmedabad

- Built and optimized a distributed data pipeline using Apache Spark (Databricks) to process cybersecurity and financial telemetry, accelerating feature engineering and model experimentation.
- Trained and validated a malicious packet detection model achieving 94% accuracy for Connected Autonomous Vehicles, improving threat response and system reliability.
- Developed LSTM-based stock forecasting models, achieving 12% RMSE reduction, and delivered insights via Tableau and Matplotlib dashboards to support financial planning decisions.

SDE Intern

Identiq Infotech

June 2022 – July 2022

Surat

- Designed and implemented a log processing pipeline handling 800–1,200 log events/sec using Apache Kafka and Flask microservices, routing enriched logs into Elasticsearch + Kibana for centralized monitoring and reducing manual log lookup time by 25%.
- Deployed LSTM and Isolation Forest models for anomaly detection on system logs, achieving 82% detection precision across multiple log categories, and served inference via FastAPI microservices to support early incident flagging.
- Built scalable ETL workflows to load structured log data into a central analytics warehouse, and created Tableau dashboards to visualize anomaly patterns, performance trends, and operational KPIs for engineering stakeholders.

Projects

Second-Hand Trading Platform

September 2024

- Led system design and product roadmap for a campus-focused trading marketplace; guided MVP features through user interviews and iterative prototyping.
- Developed full-stack application using React, Node.js, and MySQL with RESTful APIs, JWT/OAuth2 authentication, and modular backend service layers.
- Engineered real-time transactions and payments (WebSockets + Stripe) and deployed on AWS with Kubernetes and automated CI/CD, enhancing platform scalability, reliability, and user trust.

Meeting Analytics System

December 2024

- Built a multi-modal platform that converts live or uploaded audio into searchable transcripts, topic clusters, and action-item summaries using Whisper ASR, BERTopic, and LLM-based summarization pipelines.
- Developed scalable FastAPI microservices with WebSocket streaming and PostgreSQL storage to enable real-time transcript updates, speaker analytics, and conversation insights.
- Containerized and deployed on AWS with Docker + CI/CD, and built a Streamlit dashboard to review summaries and insights.

Clinical Text Analysis for Risk Prediction

March 2025

- Built a clinical risk prediction model using BioBERT embeddings + XGBoost, achieving AUC-ROC 0.89 on MIMIC-IV discharge notes to support early-intervention decisions.
- Preprocessed and reduced document length by 65% using scispaCy + embedding compression, improving model runtime and memory efficiency.
- Applied SHAP and ROUGE-based interpretability to communicate model reasoning to clinical stakeholders and ensure actionable, trust-aligned outputs.

Research Experience

Blockchain and Quantum-based Collaborative Communication Framework for Telehealth

<https://ieeexplore.ieee.org/document/10880726>

ML based Secure Communication Framework for Connected Autonomous Vehicles

<https://ieeexplore.ieee.org/document/10465158>

Technical Skills

Technical Skills: Python, Java, C++, JavaScript, TypeScript, React.js, Node.js, Streamlit, Material UI, Flask, HTML/CSS, RESTful APIs, MySQL, PostgreSQL, MongoDB, Neo4j, Apache Spark, PySpark, Kafka, Apache Airflow

AI/ML & Data Science: Machine Learning, Deep Learning, Generative AI, NLP, Computer Vision, TensorFlow, PyTorch, Scikit-learn, Pandas, OpenCV, Matplotlib, Tableau, Power BI

Cloud & DevOps: AWS, Azure, Docker, Kubernetes, Jenkins, Git, GitHub Actions, Linux, Jira